

1 One-line project summary (start with this)

"I built a console-based Bus Reservation and Ticketing System in Java that allows seat management, passenger booking, billing with discounts, and transaction tracking using structured input validation and in-memory data storage."

2 Problem statement (what you were solving)

Manual ticket booking is error-prone and slow.

This project simulates a real-world bus reservation system where users can manage destinations, seat availability, passenger bookings, billing, and payment status through a menu-driven interface.

3 High-level system flow (VERY IMPORTANT)

Explain this in **steps**:

◆ Step 1: Authentication

- Simple login using username and password
- Prevents unauthorized access

"This simulates role-based access at a basic level."

◆ Step 2: Main Menu (Core Modules)

The system has **5 main modules**:

1. **Destination Management**
2. **Passenger Booking**
3. **Billing & Payment**
4. **View Passenger Details**
5. **Exit**

4 Module-wise explanation

◆ 1. Destination Module

- Displays available destinations
- Shows:
 - Fare per destination
 - Remaining seat count
- Seats are initialized dynamically

"Seat availability updates in real time after each booking."

◆ 2. Passenger Booking Module

This is the **core logic**.

What happens here:

- Passenger name validation (no duplicates)
- Destination selection with validation
- Seat availability check
- Passenger count input

- Discount handling (PWD / student / senior – 20%)
- Fare calculation

Business logic:

Total Fare =

(Regular passengers × fare)
+ (Discounted passengers × (fare – 20%))

"This ensures accurate billing while enforcing seat limits."

◆ **3. Billing & Payment Module**

- Search passenger by name
- Prevent double payment
- Accept payment
- Compute change
- Mark passenger as PAID

"Once paid, the passenger status is locked to avoid duplicate billing."

◆ **4. View Module**

- Search passenger details
- Displays:
 - Destination
 - Passenger count
 - Discount count
 - Total fare
 - Payment status
 - Amount paid and change (if applicable)

"This acts like a transaction history lookup."

◆ **5. Exit Module**

- Gracefully exits the system
- Confirms continuation